# The Lunar Outer Shell as a Weakly Conducting, Large–Skin-Depth Medium at Extremely Low Frequencies

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**Abstract**

We present a quantitative, observation-based assessment of electromagnetic propagation in the outer layers of the Moon at extremely low frequencies (ELF). Continuous electrical-conductivity profiles—anchored to Apollo-era and modern magnetic-sounding inversions—are used to evaluate skin depth, loss tangent, circumferential path attenuation, and global-mode quality factors. At 1–30 Hz, literature-bracketed outer-shell conductivities yield loss tangents $\tan\delta = \sigma/(\omega\varepsilon) \gg 1$: conduction current dominates displacement current. The outer shell is therefore a *weakly conducting medium with large skin depth*, not a low-loss dielectric. Without a conducting ionosphere there is no closed Earth-like Schumann cavity. Even granting a hypothetical 100 km ionosphere, impedance-method cavity quality factors remain $Q \lesssim 2$ across named profiles, and a 20 000-draw Monte Carlo over the conductivity envelope finds median $Q \approx 0.78$, 95th percentile $Q \approx 1.85$, and maximum $Q \approx 2.0$ (100% of draws have $Q<5$). Nearside–farside and Procellarum KREEP Terrane contrasts alter path attenuation but do not create a high-Q resonator. The method supplies a reproducible framework for subsurface ELF assessment on the Moon and other airless bodies.

## 1 Introduction

Assessing whether a planetary interior can support low-attenuation ELF propagation is a recurring problem in planetary geophysics. The Moon is a clean test case: it lacks liquid water and a dense atmosphere, and electromagnetic sounding shows its outer layers to be highly resistive relative to terrestrial crust. Prior qualitative arguments have sometimes described the outer shell as a "low-loss dielectric." That terminology is tested here against published conductivity structure and explicit modal quality-factor estimates.

The method is deliberately simple:

1. Assemble continuous $\sigma(r)$ profiles from published magnetic-sounding inversions and literature-bracketed envelopes.
2. Compute skin depth, loss tangent, and path attenuation at ELF.
3. Estimate global-mode quality factors for an open Moon and for a hypothetical closed cavity with an artificial ionosphere.
4. Propagate uncertainty with Monte Carlo draws and assess nearside–farside / PKT lateral contrasts.

## 2 Conductivity Structure of the Lunar Outer Shell

Electrical conductivity of the lunar interior is inferred from magnetic induction in time-varying external fields, using dual-spacecraft measurements from Apollo surface magnetometers and lunar orbiters (e.g., Explorer 35) and, more recently, orbital magnetometer inversions. Re-analyses confirm that conductivity is extremely low in the outer few hundred kilometers and rises smoothly with depth as temperature increases [Sonett et al., 1971; Dyal and Parkin, 1971; Mittelholz et al., 2021; Grimm, 2023].

A preferred one-dimensional construction for this study combines:

- a cold resistive lid consistent with Dyal and Parkin (1977) near-surface bounds ($\sigma \lesssim 10^{-8}$ S/m for depths $\lesssim$ 80 km);
- the Grimm (2023) low-frequency (LF) analytic fit valid over ∼400–1200 km, $\sigma = 1.76 \times 10^{-4} \exp(z_{km}/210)$ S/m;
- a log-linear bridge between the lid and the LF band;
- end-members: a Mittelholz-like global orbital envelope (constructed), a classical Hood-class resistive outer shell, and a Grimm high-frequency (HF) elevated envelope retained only as an upper bound (Grimm argues HF is likely biased high).

There is no sharp step-function boundary: the transition from resistive to more conductive rock is continuous. Consequently the outer shell cannot be treated as a homogeneous dielectric slab terminated by a perfect conductor.

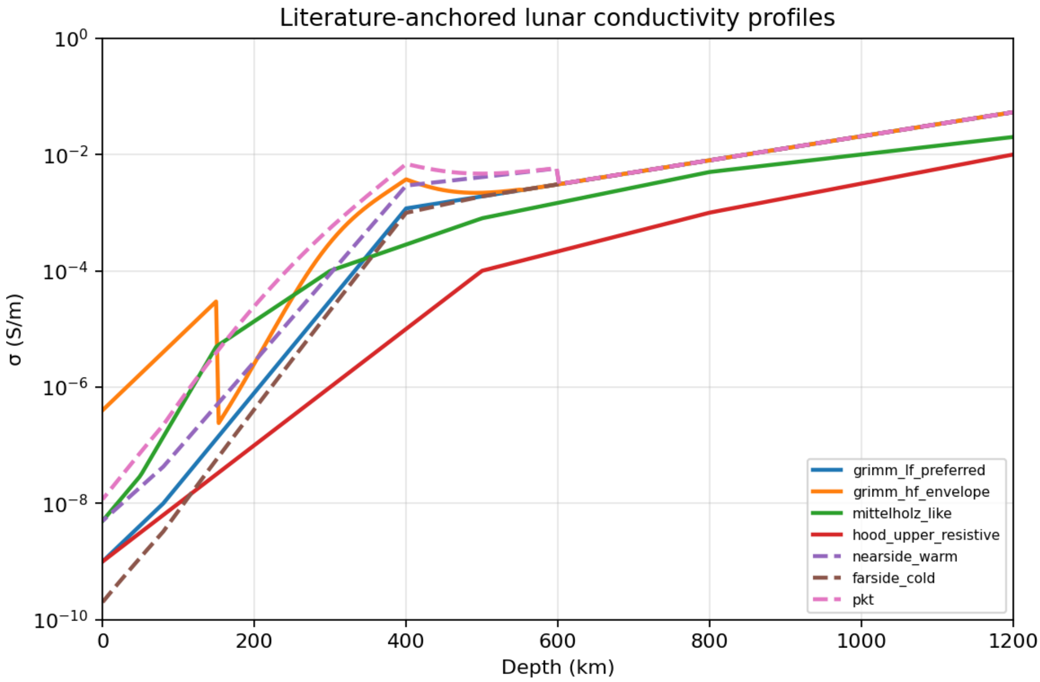


**Figure 1.** Literature-anchored lunar conductivity profiles used in this study. Solid curves: global 1-D constructions (Grimm LF preferred, HF envelope, Mittelholz-like, Hood-class resistive). Dashed: regional nearside / farside / Procellarum KREEP Terrane (PKT) variants.

## 3 Electromagnetic Characteristics at ELF

At frequencies of a few to a few tens of hertz, the electromagnetic skin depth in the good-conductor limit,

$$\delta = \sqrt{(2)/(\omega\mu\sigma)}\ ,$$

remains on the order of tens to thousands of kilometers within the outer resistive region. For $f = 10$ Hz and $\sigma = 10^{-7}$ S/m with $\mu = \mu_0$, $\delta \approx 500$ km.

The relative magnitude of conduction to displacement current is the loss tangent

$$\tan\delta = (\sigma)/(\omega\varepsilon)\ .$$

For lunar rock with $\varepsilon_r \sim$ 4–7, $\omega\varepsilon$ is of order $10^{-9}$ S/m at 10 Hz. Thus for $\sigma=10^{-7}$ S/m, $\tan\delta \sim$ 30–50 $\gg$ 1: conduction current dominates. A true low-loss dielectric would require $\tan\delta \ll 1$, i.e. $\sigma \ll 10^{-9}$ S/m at 10 Hz.

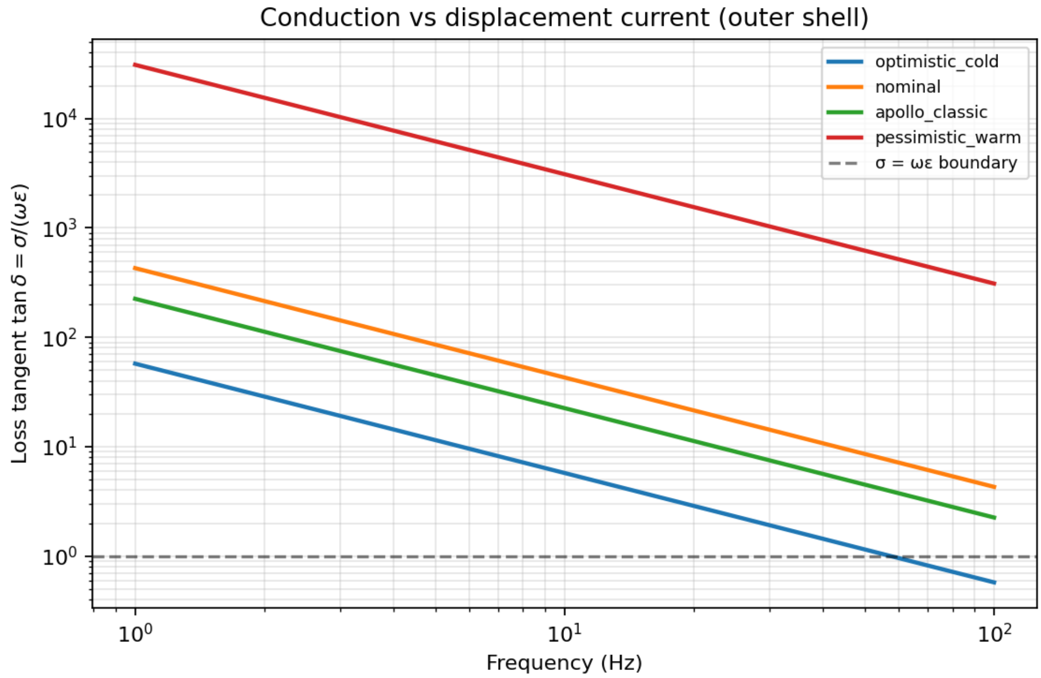


**Figure 2.** Loss tangent $\tan\delta=\sigma/(\omega\varepsilon)$ for outer-shell (0–300 km) log-mean conductivity of the synthetic profile suite. Values $\gg$ 1 indicate conduction-dominated (weakly conducting) behavior across 1–100 Hz.

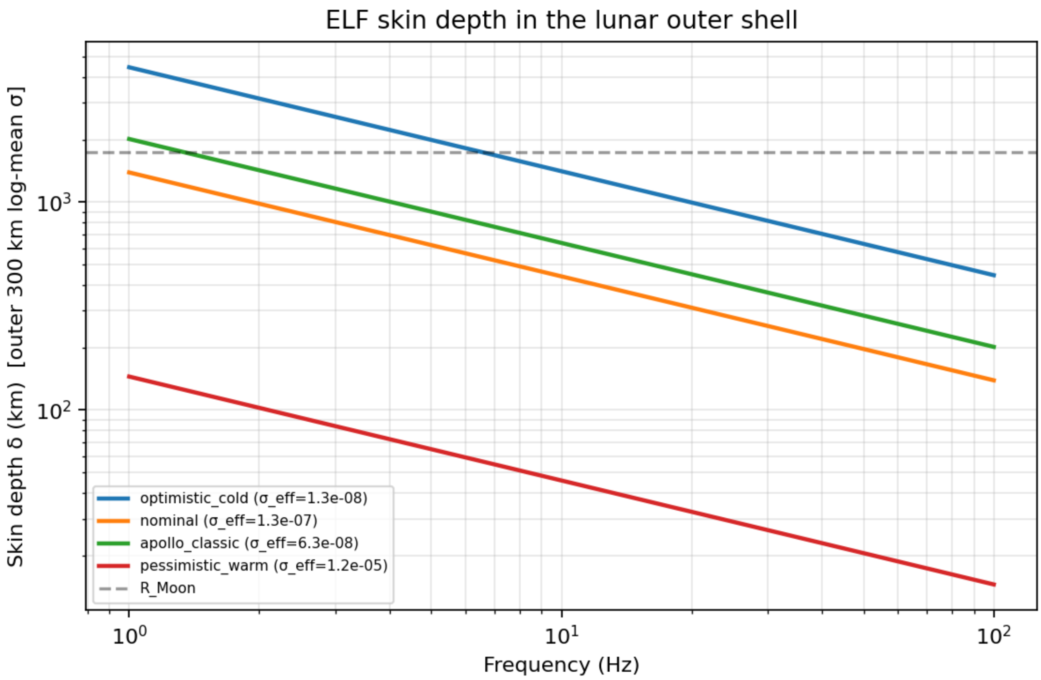


**Figure 3.** ELF skin depth versus frequency for outer-shell effective conductivity. Large $\delta$ implies low attenuation per kilometer, but a lunar circumference still spans many skin depths for nominal profiles.

Table 1 summarizes outer-shell metrics for literature constructions at 10 Hz. Half-circumference field attenuations of tens to hundreds of decibels show that shell-guided global standing waves are not viable for nominal and midmantle-resolved profiles.

| **Profile** | $\sigma_{eff}$ **(S/m)** | **δ (km)** | **tan δ** | $A_{1/2circ}$ **(dB)** |
|---|---|---|---|---|
| Grimm LF preferred | $1.4\times 10^{-7}$ | 429 | 45 | 111 |
| Mittelholz-like | $1.6\times 10^{-6}$ | 125 | 530 | 379 |
| Hood-class resistive | $3.0\times 10^{-8}$ | 917 | 9.9 | 52 |
| Grimm HF envelope | $5.7\times 10^{-6}$ | 67 | 1850 | 710 |

**Table 1.** Outer-shell (0–300 km log-mean) metrics at 10 Hz for literature-anchored profiles.

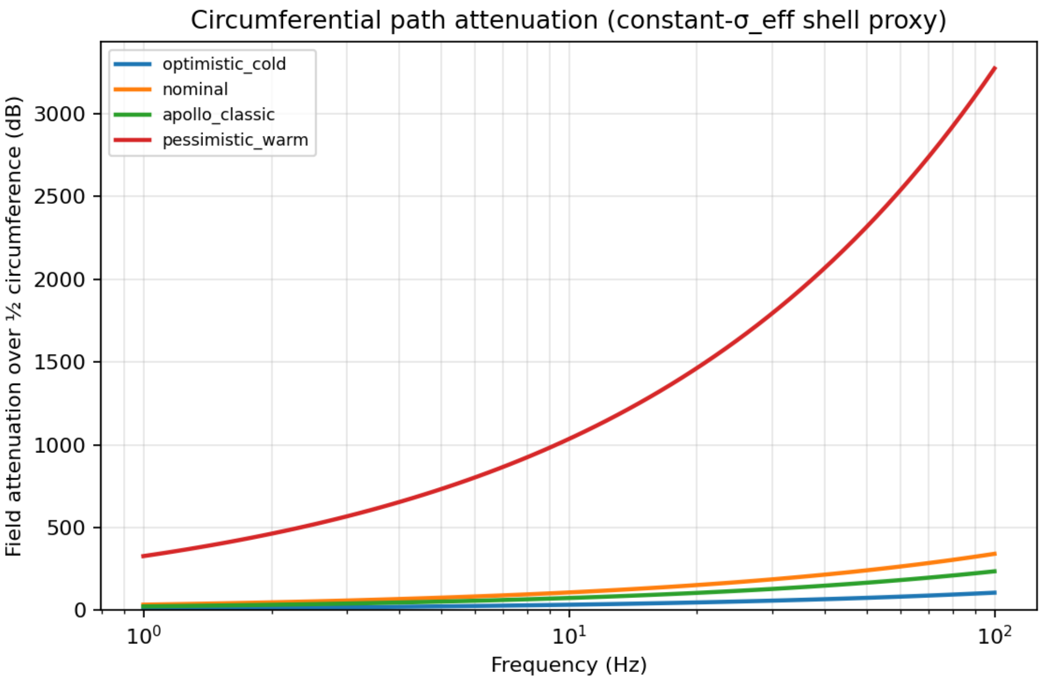


**Figure 4.** Circumferential path attenuation (half lunar circumference) for the synthetic profile suite, illustrating strong global-shell damping at ELF for all but the most extremely resistive end-members.

## 4 Global Resonances and Quality Factors

Ideal closed-cavity Schumann frequencies for a sphere of lunar radius R are $f_n=(c/2\pi R)\sqrt{n(n+1)}$, so $f_1\approx 38.8$ Hz. The physical Moon has no dense conducting ionosphere to form the upper wall of an Earth-like cavity (Model A). Global high-Q standing waves of the classic Schumann type are therefore not supported by geometry alone.

To bound residual “would-be” cavity quality even under favorable assumptions, we adopt a hypothetical closed cavity (Model B) with a conducting ionosphere at height h=100 km. Planetary surface impedance $Z_g(\omega)$ is computed from a layered magnetotelluric-style recursion through σ(r); ionospheric impedance $Z_i$ is a uniform half-space proxy. The cavity quality factor is estimated as

$$Q\approx(\omega\mu_0 h)/(2\, Re(Z_g+Z_i))\ .$$

An Earth validation stack (Model C) recovers Q∼ 4–6 for the first three multipoles—order-of-magnitude agreement with terrestrial Schumann Q of a few to ten.

| Profile | Q (n=1) | Q (n=2) | Q (n=3) |
|---|---|---|---|
| Grimm LF preferred | 0.78 | 0.64 | 0.45 |
| Mittelholz-like | 1.08 | 1.08 | 0.99 |
| Hood-class resistive | 0.49 | 0.34 | 0.20 |
| Grimm HF envelope | 1.11 | 1.32 | 1.47 |
| Nearside warm | 0.78 | 0.69 | 0.57 |
| Farside cold | 0.83 | 0.66 | 0.43 |
| PKT | 0.94 | 0.93 | 0.86 |

**Table 2.** Model B cavity Q (artificial 100 km ionosphere) at ideal multipole frequencies for literature and regional columns. Primary metric: impedance method (Eq. 3).

Ringing times tau≈ Q/(π f_1) are a few milliseconds for Q∼ 1 at f_1∼ 39 Hz—orders of magnitude too short for a sustained global ring.

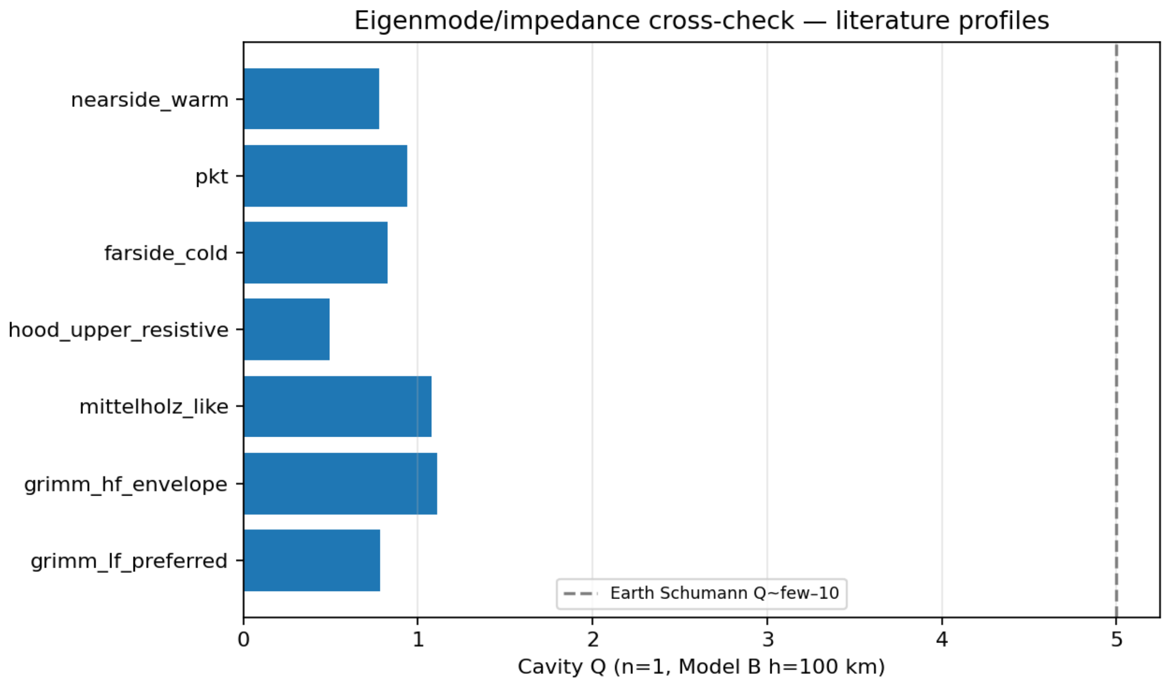


**Figure 5.** Fundamental-mode cavity Q for literature and regional profiles under Model B (h=100 km). All values remain O(1), far below high-Q resonator behavior.

### *4.1 Monte Carlo conductivity envelope*

To avoid reliance on a single σ(r), we draw 20 000 synthetic profiles with surface-to-deep conductivity hinges spanning optimistic to pessimistic literature bounds, and recompute Model B Q for n=1. Results:

- median Q=0.78;
- 5th / 95th percentiles: 0.30 / 1.85;
- maximum Q=2.01;
- **100% of draws have Q<5**; 65% have Q<1.

No high-Q island appears in the envelope.

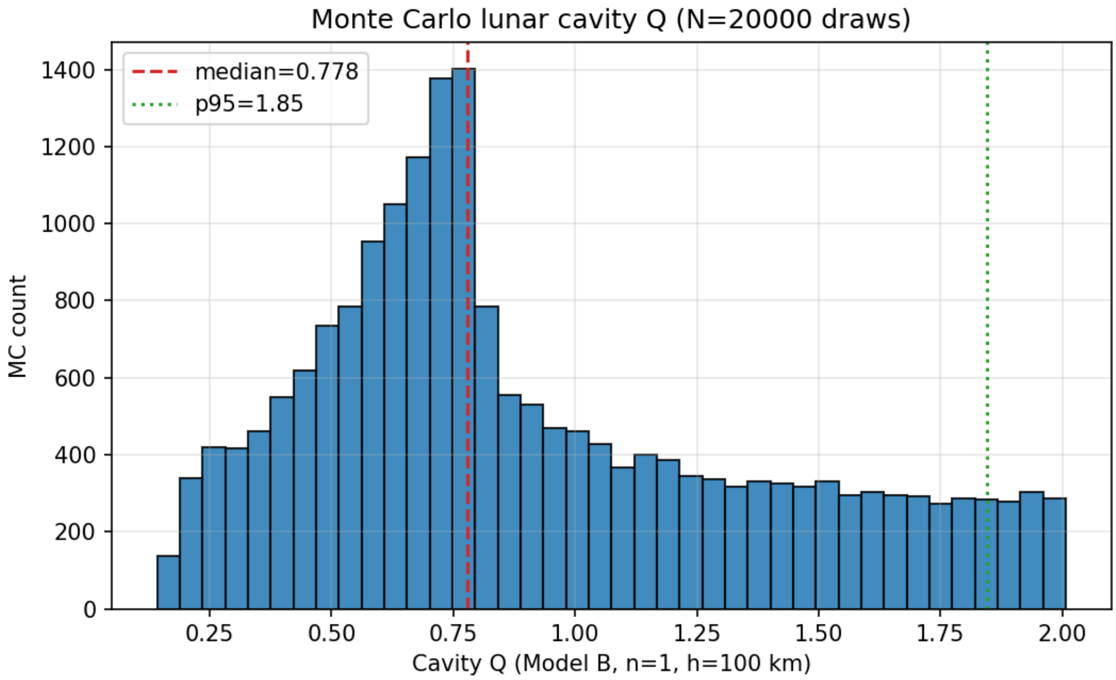


**Figure 6.** Monte Carlo distribution of Model B fundamental-mode Q (20 000 draws). Median and 95th percentile are marked; the entire support lies well below terrestrial Schumann Q.

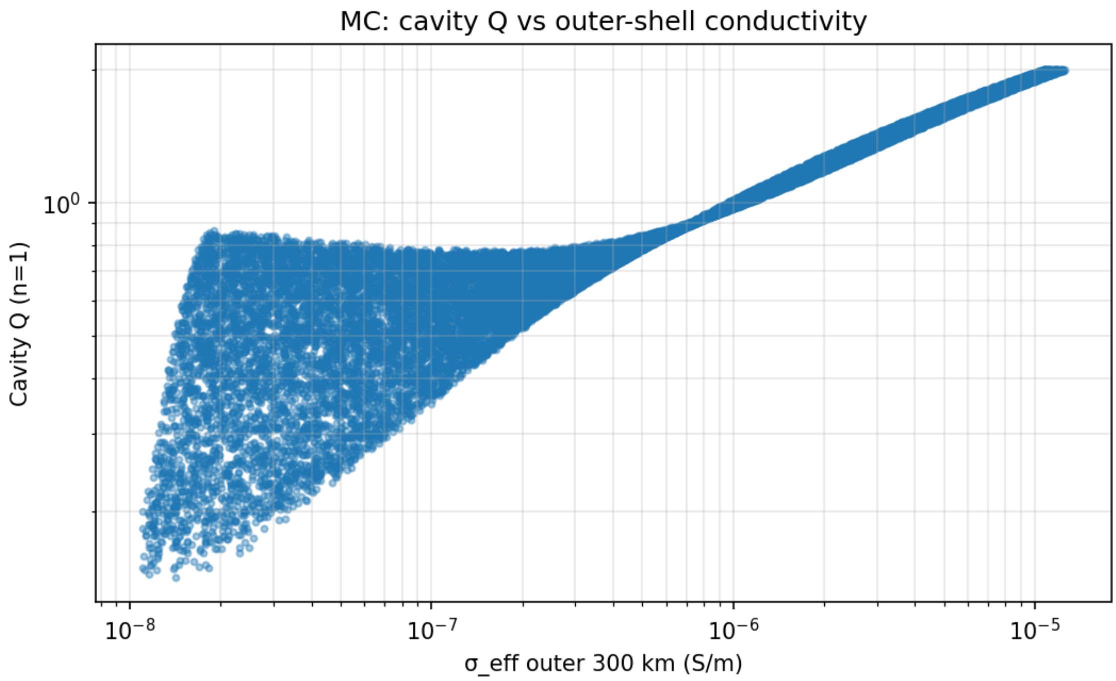


**Figure 7.** Monte Carlo Q versus outer-shell effective conductivity. More conductive walls lower Re(Z_g) and can raise Q slightly, but values remain $\lesssim 2$ across the envelope.

## 5 Lateral Heterogeneity

One-dimensional inversions miss nearside–farside structure and the Procellarum KREEP Terrane. Regional column variants (warmer nearside, colder farside, PKT-boosted upper mantle) change local surface impedance and shell-path attenuation. Two equal-area hemisphere combinations yield effective cavity Q∼ 0.8–0.9—still O(1). Piecewise great-circle paths at 10 Hz give ∼36 dB (farside) to ∼268 dB (PKT) of attenuation over 90°, and ∼137 dB on a mixed path through PKT. Lateral contrast modulates transmission but does not produce a high-Q global mode.

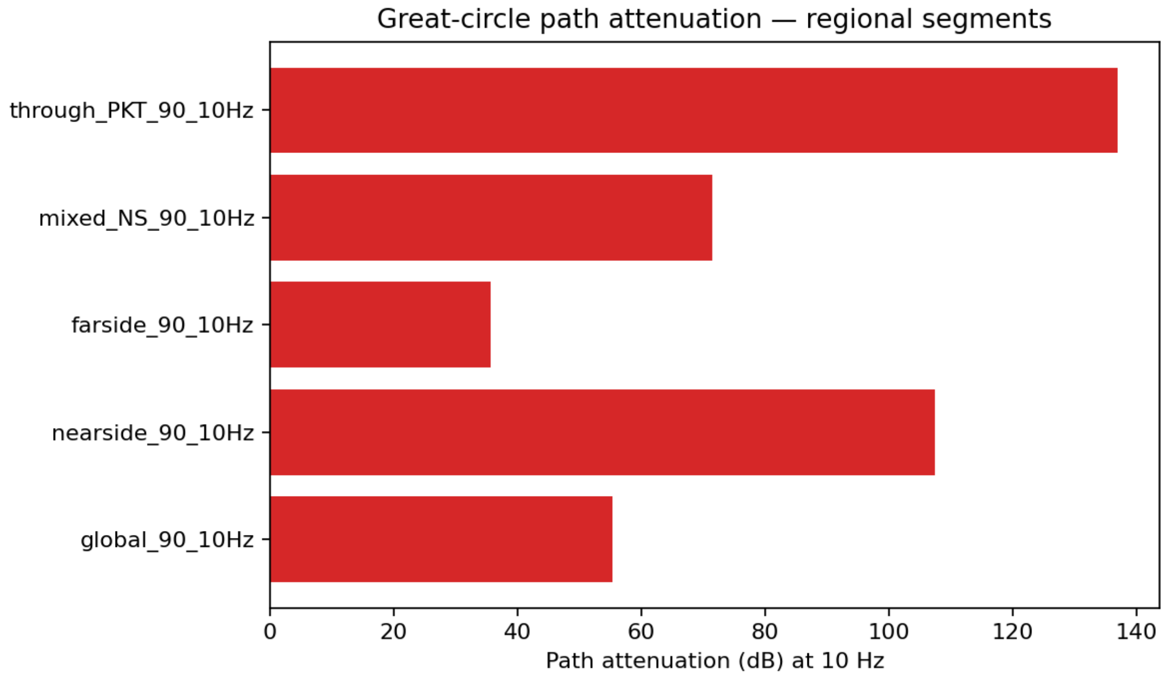


**Figure 8.** Great-circle path attenuation at 10 Hz for global, nearside, farside, mixed, and PKT-crossing segments (90° total arc unless noted in labels).

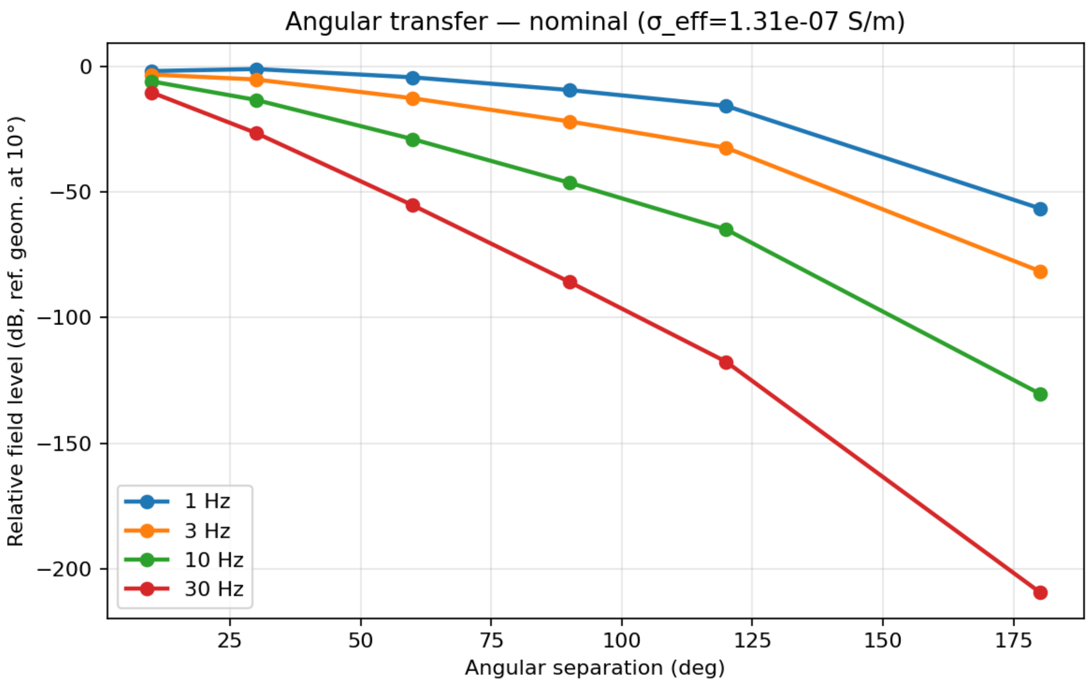


**Figure 9.** Source–receiver transfer proxy versus angular separation for a nominal outer-shell conductivity (path attenuation plus geometric spreading).

## 6 Limitations and Open Questions

- Existing conductivity profiles have limited vertical resolution in the uppermost 200–300 km; future magnetotelluric-style measurements would tighten the lid.
- The Mittelholz-like curve is a constructed global envelope consistent with orbital-class midmantle conductivities, not a point digitization of a single published figure.
- Model B ionospheres are artificial proxies for bounding Q; the physical Moon lacks this wall.
- Full three-dimensional Maxwell solutions with continuous lateral structure remain future work; the regional path and two-hemisphere estimates bound the effect of first-order asymmetry.
- Displacement current and vacuum radiation for the open Moon are not treated as a closed eigenvalue problem; Model A is geometric (no cavity), not a leaky-mode catalog.

## 7 Conclusion

Continuous electrical-conductivity profiles derived from Apollo and orbital magnetic sounding, together with literature-bracketed envelopes and a 20 000-draw Monte Carlo experiment, show that the outer several hundred kilometers of the Moon form a **weakly conducting, large-skin-depth medium** at ELF—not a low-loss dielectric cavity wall. Loss tangents satisfy $\tan\delta \gg 1$ at 1–30 Hz. The absence of a conducting ionosphere precludes a closed Schumann-type cavity; even with a hypothetical ionosphere, cavity quality factors remain $Q \lesssim 2$ across the explored envelope. Lateral nearside–farside and PKT structure modulates path attenuation but does not enable high-Q global resonances. The same observation-based template applies to other airless bodies.


## Acknowledgments

Quantitative results were produced with the open `lunar-elf` modeling package (layered surface-impedance cavity Q, path integrals, Monte Carlo, and regional path experiments). Reproducibility scripts and figures are archived with the project.


## References


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*Corresponding code and full numerical tables:* `Projects/lunar-elf/` *(*`paper/QUANTITATIVE_RESULTS.md`*,* `paper/CAMPAIGN_REPORT.md`*,* `paper/OPTIONAL_REPORT.md`*).*